LAB 6

AIM:	To implement the T-test model
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Steps.:

1. Import the required packages
2. Create the two variable x & y
3. Assign the values to the 2 variable
4. Find the standardization of it
5. Print the result
6. End the result

Program.:

```
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from scipy import stats

np.random.seed(42)
a = np.random.normal(80, 8, 30)
b = np.random.normal(75, 10, 30)

t_stat, p_val = stats.ttest_ind(a, b)

print(f"Analysis Results:")
print(f"{'-'*20}")
print(f"T-statistic: {t_stat:.3f}")
print(f"P-value:     {p_val:.3f} ({'Significant' if p_val < 0.05 else 'Insignificant'})")

plt.figure(figsize=(8, 5))
sns.kdeplot(a, fill=True, color='royalblue', label='University A (Mean=80)')
sns.kdeplot(b, fill=True, color='seagreen', label='University B (Mean=75)')
```

Program:	plt.title('Technical Assessment: Score Distribution Comparison', pad=15) plt.xlabel('Assessment Score') plt.ylabel('Density') plt.legend() sns.despine() plt.tight_layout() plt.show()
Output: